

Snort

Source: <https://www.snort.org>

Snort is an open-source network intrusion detection system, that can perform real-time traffic analysis and packet logging on IP networks. Snort's rule engine enables custom rules to meet the needs of the network. Snort rules help in differentiating between normal internet activities and malicious activities.

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Structure of a Snort Rule

Rule Header

Rule options

Basic structure of Snort rules

Action	Protocol	Address	Port	Direction	Address	Port
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Structure of Snort rule header

1.Rule Header

Rule Actions

Action	Description
Pass	Tells Snort to ignore the packet
Log	To log a packet
Alert	To create an alert and then to activate another rule for checking more conditions
Dynamic	Dynamic action rules are invoked by other rules using the "activate" action
User Defined Actions	You can define your actions

Protocols

Rule	Description
alert icmp any any -> any any (msg: "Ping with TTL=100"; \ ttl: 100;)	Shows on which type of packet the rule will be applied

The Direction Operator

log !<Target Ip1>/24 any <> <Target Ip2>/24 23	<> symbol shows that the rule will be applied to packets traveling on either direction.
alert tcp any any -> <Target Ip>/24 any (flags: A; \ ack: 0; msg: "TCP ping detected");	-> shows that address and port numbers on the left-hand side of the direction field are the sources of the packet while those on the right-hand side are the destination

Port Numbers

Port ranges

alert udp any 1024:2048 -> any any (msg: "UDP ports");	Use a range of ports instead of only one port in the port field
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Negation symbol

log udp any !53 -> any any log udp	Using the negation symbol (!) with port numbers to exclude a port or a range of ports
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Addresses

alert tcp any any -> <Target Ip>/32 80 (msg: "TTL=100"; \ ttl: 100;)	Two address parts in a Snort rule are used to check the source and the destination of the packet
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Address exclusion

alert icmp ![<Ip range>] any -> any any \ (msg: "Ping with TTL=100"; ttl: 100;)

Snort provides a mechanism to exclude addresses using the negation symbol an exclamation point.

Address lists

alert icmp ![<Ip range> /24, <Ip range>/24] any -> any \ any (msg: "Ping with TTL=100"; ttl: 100;)

Specifies a list of addresses in a Snort rule

2. Rule Options

Option	Description
ACK keyword	
alert tcp any any -> <IP range> any (flags: A; \ ack: 0; msg: "TCP ping detected");	Alerts when you receive a TCP packet with the ACK flag set and it contains a value of 0
classtype Keyword	
alert udp any any -> <IP range> 6838 (msg:"DoS"; \ content: "server"; classtype:DoS;)	Do not crack a hash, instead print candidates only
content Keyword	Compare hashlist with potfile; show cracked hashes
alert tcp <IP range> any -> <IP range> any \ (content: "GET"; msg: "GET matched");	Compare hashlist with potfile; show uncracked hashes
offset Keyword	
alert tcp <IP range> any -> any any \ (content: "HTTP"; offset: <no of bytes>; msg: "HTTP matched");	Starts searching for the word "HTTP" after the specified number of bytes from the start of the data
depth Keyword	
alert tcp <IP range>any -> any any (content: \ "HTTP"; offset: <no of bytes>; depth: <no of chars>; msg: "HTTP matched");	Search the word "HTTP" between the specified range of the data part of the TCP packet
content-list Keyword	
alert ip any any -> <IP range> any (content-list: \ "yourtext"; msg: "Text matched");	Search the given strings in the data portion of all packets matching the rule criteria
dsize Keyword	
alert ip any any -> <IP range> any (dsize: > 6000; \ msg: "Large size IP packet detected");	Alerts if the data size of an IP packet is larger than 6000 bytes
flags Keyword	
alert tcp any any -> <IP range> any (flags: SF; \ msg: "SYNC-FIN packet detected");	Alerts any scan attempt using SYN-FIN TCP packets

fragbits Keyword

alert icmp any any -> <IP range> any (fragbits: D; \ msg: "Don't Fragment bit set");

Alerts if the DF bit is set in an ICMP packet

icmp_id Keyword

alert icmp any any -> any any icmp_id: 100; \ msg: "ICMP ID=100");

Verifies if the ICMP ID field in the ICMP header is equal to 100

icmp_seq Keyword

alert icmp any any -> any any (icmp_seq: 100; \ msg: "ICMP Sequence=100");

Alerts if it finds a sequence number of 100

itype Keyword

alert icmp any any -> any any itype: 4; \ msg: "ICMP Source Quench Message received");

Alerts if it detects a source quench message

icode Keyword

alert icmp any any -> any any itype: 5; \ icode: 1; msg: "ICMP ID=100");

Alerts if host redirect ICMP packets are received

ipopts Keyword

alert ip any any -> any any (ipopts: lsrr; \ msg: "Loose source routing attempt");

Detects and alerts Loose Source Routing attempts

ip_proto Keyword

alert ip any any -> any any (ip_proto: ipip; \ msg: "IP-IP tunneling detected");

Alerts if IPIP protocol is used by data packets

logto Keyword

alert icmp any any -> any any (logto:<filename>; ttl: 100;)

Records all ICMP packets having TTL value equal to 100 to the specified file

priority Keyword

alert ip any any -> any any (ipopts: lsrr; \ msg: "Loose source routing attempt"; priority: 10;)

Assigns a priority to a rule

react Keyword

alert tcp <IP range> any -> any 80 (msg: "Outgoing \ HTTP connection"; react: block;)

Terminate a session to block some sites or services

reference Keyword	
<pre>alert udp \$EXTERNAL_NET any -> \$HOME_NET 1900 \ (msg:"MISC UPNP malformed advertisement"; \ content:"NOTIFY * "; nocase; classtype:misc-attack; \ reference:cve,CAN-2001-0876; reference:cve, \ CAN-2001-0877; sid:1384; rev:2;)</pre>	Alert shows a reference to find more information about this alert. Adds a reference to information present on other systems available on the Internet
resp Keyword	
<pre>alert tcp any any -> <IP range> 8080 (resp: rst_snd;)</pre>	Sends a TCP Reset packet to the sender if an attempt is made to connect to TCP port 8080 on the local network
rpc Keyword	
<pre>alert ip any any -> 192.168.1.0/24 any (rpc: 10000,*3; \ msg: "RPC request to local network");)</pre>	Detects and alerts RPC based requests
sameip Keyword	
<pre>alert ip any any -> <IP range> any (msg: "Same IP"; \ sameip;)</pre>	Alerts if source and destination IP addresses are same
session Keyword	
<pre>log tcp any any -> <IP range> 110 (session: printable;)</pre>	Rule logs all printable data from POP3 sessions
tag Keyword	
<pre>alert tcp <IP range> 23 -> any any \ (content: "facebook"; msg: "Detected facebook"; \ tag: session, 100, packets;)</pre>	Records 100 packets on the session after it is detected

3. Command Line Options

Syntax: snort [options]

Option	Description
-A alert-mode	Generates an alert using one of the specified alert-modes: fast, full, none, and unlock
-b	Logs packets in tcpdump format (i.e., libpcap)
--brain-host	Scrambles the networks specified in the -h (or HOME_NET) setting
-c config-file	Allows you to specify which configuration file you want to use
-C	Prints the character data found in the packet payload
-d	Displays the application layer data when in verbose or packet logging mode
-D	Turns Snort in daemon mode
-e	Displays or logs the link-layer packet headers
-F bpf-file	Reads Berkeley Packet Filters (BPF) from a bpf file
-g group	Changes the default group ID or GID under which Snort runs after initialization
-h home-net	Sets the "home network" to a specific address in CIDR format

Option	Description
-i interface	Specifies which interface Snort should listen on
-I	In alerts, displays the interface on which each packet arrived
-k checksum-mode	Controls which packet checksums Snort computes and verifies
-l logging-directory	Specifies the logging directory
-L binary-log-file	Specifies the logging directory
-m umask	Sets the file mode creation mask to the designated umask variable
-n packet-counte	Processes the given number of packets and then exits
-N	Turns off packet logging
-o	Changes the order in which the rules are applied to packets
-O	Replaces the IP addresses printed to the screen or log file with "xxx.xxx.xxx.xxx"
-p	Turns off promiscuous mode sniffing
-P snap-length	Sets the maximum packet capture length to a certain size
-q	Tells Snort to run quietly. Does not display banner and initialization information
-r tcpdump-file	Processes a tcpdump-formatted file
-s	Sends alert messages to a syslog server
-S variable=value	Sets the variable name <variable> to the value <value>
-t chroot	Changes Snort's root directory to chroot after initialization
-T	Starts Snort in self-test mode.
-u user	Changes the default user ID or UID under which Snort runs after initialization
-U	Forces the timestamp in all logs to be in UTC format
-v	Prints all packets to the console
-V	Displays the Snort version number and then exits
-X	Displays raw packet data starting at the link layer
-Y	Includes the year in all alerts and log files
-z	Enables the stream4 preprocessor
-?	Lists all switches and options and then exits

4. Sniffer Mode

Sniff packets and send to standard output as a dump file	
Options	Description
-v (verbose)	Display output on the screen
-e	Display link-layer headers
-d	Display packet data payload
-x	Display full packet with headers in HEX format

5. Packet Logger Mode

Input output to a log file	
Options	Description
-r	Use to read back the log file content using snort
-l (directory name)	Log to a directory as a tcpdump file format
-k (ASCII)	Display output as ASCII format

6. NIDS Mode

Use the specified file as config file and apply rules to process captured packets	
Options	Description
-c	Define configuration file path
-T	Use to test the configuration file including rules